Q20 Deeper Project Suggestions

1. **Lineage audit:** implement a 2-layer CNN, RNN, and Transformer in the same notebook with a shared training loop; tabulate which code lines are architecture-specific vs. shared. Quantify: what fraction of the *code* is the skeleton?
2. **Mixer swap experiment:** train identical-parameter small models where only the sequence mixer differs (attention vs. conv vs. recurrence). Plot loss curves and training wall-clock; measure parallelism sensitivity by varying sequence length.
3. **Transfer-of-intuition test:** take three classic vision diagnostics (LR-range test, grad-norm monitoring, overfit-one-batch) and systematically apply them to small LM training; document which transfer and which mislead.
4. **Post-training on a non-LLM:** apply DPO-style preference optimization to a small classifier's logits and characterize what breaks; write up whether "post-training" is inherently generative.